

 PROJECT OVERVIEW

Detecting Concept & Non-IID Federated Learning Environment

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01 What is Federated Learning?

Federated Learning (FL) allows multiple organisations to train a shared AI model together — without sharing their raw data. Each organisation trains locally on their own data and only sends model weights (patterns) to a central server.



02 The Two Problems



Non-IID Data

Each organisation always has different data. Bank A has rich customers, Bank B has students. This is permanent and expected — it never changes.



Concept Drift

An organisation's data suddenly changes over time. Fraud patterns evolve. New virus appears. This is unexpected and causes the model to fail.

03

The Research Gap

When the server receives a bad update from a client, it cannot tell whether:



Non-IID?

This client was always different — it is just unique. Normal! Ignore it.



Concept Drift?

This client's data just changed — something happened! Act now!

"Existing research addresses Non-IID and Concept Drift separately. No work uses a client's own gradient history to distinguish between the two."

04

Research Question

"Can cosine similarity detect concept drift in a Non-IID Federated Learning environment?"

05

Proposed Solution — Cosine Similarity

Compare each client's current gradients to its own historical gradients. The pattern tells us what happened:

✔️ NON-IID CLIENT — Stable

Day 1	0.3
Day 10	0.3
Day 20	0.3
Day 30	0.3

→ Always low, always stable = Normal!

⚠️ DRIFTING CLIENT — Sudden Drop

Day 1	0.8
Day 10	0.8
Day 20	0.8
Day 26	0.1

→ Sudden drop = DRIFT detected!

06

3 Models Compared

Model	What it does	Normal	Non-IID	Drift	Both
Simple ML	No FL — basic model	78%	55%	41%	38%
FedAvg	Standard FL — blind to drift	85%	61%	52%	44%
Our Method 	Cosine similarity detects drift	87%	74%	76%	71%

* Numbers are estimated targets — actual results from experiments

07 Tools & Technologies



Flower (flwr)

FL framework — simulates clients and server



PyTorch

Deep learning — model training



CIFAR-10

Standard benchmark dataset — free, public



Cosine Similarity

Main detection tool — PyTorch built-in